

Wenyu Chiou

AI Agent Infrastructure & Evaluation Engineer

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I build AI-agent and research infrastructure — MCP research pipelines, multi-agent orchestration, and portable agent harnesses — and governance frameworks that keep LLM-agent simulations of human–environment systems stable and verifiable.

— SELECTED SYSTEMS

research-hub MAINTAINED OPEN-SOURCE TOOL

An AI-operable research workspace: gives an AI assistant a CLI, MCP server, REST API, and dashboard to run repeatable literature workflows — search papers across arXiv, Semantic Scholar, PubMed, and CrossRef, ingest them into Zotero, sync notes to Obsidian, and verify NotebookLM briefs against their sources. Sole designer and maintainer end to end.

PyPI `research-hub-pipeline v1.1.1` · listed in Awesome MCP Servers · CI across 3 OS × 4 Python versions · CLI + MCP + REST + dashboard

ai-research-skills MAINTAINED OPEN-SOURCE TOOL

A catalog of 15 research-workflow skills, packaged as a five-plugin marketplace with one-command install. It opens with a 3-gate decision dossier — is the research gap open, a real contribution, and feasible? — so the downstream stages (design, build, drafting, reviewer response) only run on a vetted topic. The same skill files load across Codex, Cursor, Gemini, and generic API clients.

15 skills · 5-plugin marketplace · cross-runtime SKILL.md contract · MIT

fable-method-harness MAINTAINED OPEN-SOURCE TOOL

A portable operating harness for AI-assisted work: model / cost routing, honest-failure discipline, and a per-step audit trail. Measured against a benchmark suite rather than asserted; verified across Claude and Codex runtimes with reserved slots for others.

validation CI · benchmark + distillation harness · routing playbook · audit-trail memory

agent-collab-skills & codex-delegate MAINTAINED OPEN-SOURCE TOOLS

Multi-agent orchestration (task splitter, output reconciler, adversarial debate, shared memory, pre-merge acceptance gate) plus cross-platform delegation wrappers whose contracts verify change from git state, not the delegate's own report. A regression test pins a real upstream stdin hang and fails without the fix.

test suites + CI on Ubuntu and Windows · contract + failure-mode tests · composable

WAGF — Water Agent Governance Framework IN PREPARATION FOR RELEASE

A governance framework for LLM agents in human–water simulation: agent decisions are constrained by domain rules and behavioral theory, with retry feedback and audit logs, so that the coupled behavioral simulation's outcomes stay stable and physically sensible instead of drifting into implausible states.

domain-rule + behavioral-theory constraints · retry feedback · audit logs

— COMMUNITY & EDUCATION

awesome-agentic-ai-zh — a trilingual (zh-TW / zh-Hans / English) 7-stage learning roadmap for building agentic AI, with 240+ curated resources from LLM basics to production multi-agent systems, CI-enforced locale parity, and external contributors. 4.5k+ GitHub stars, 588 forks (July 2026).

— OPEN-SOURCE CONTRIBUTIONS

- 10 merged pull requests in third-party open-source projects, including a root-caused engine-selection bug fix with a regression test that fails on main (BuilderIO/agent-native ×4) and a Windows file-permissions security fix (langchain-ai/openwiki).

- Taiwan-market data features across three merged PRs (ZhuLinsen/daily_stock_analysis ×3, re-scoped through maintainer review); pure-bash UTF-8 / East-Asian-width display fix (Nanako0129/coralline); documentation fix (NVIDIA/skills).

— PUBLICATION

Yang, Y. C. E., & Chiou, W. (2026). Leveraging Large Language Models for Agent-Based Simulation of Human–Water System Interactions. *Water Resources Research*, 62(6), e2025WR042111. doi.org/10.1029/2025WR042111 Published · second author of two.

— DOMAIN & RISK ANALYTICS

FLOODABM — 52,141-household coupled agent-based × catastrophe flood model with NFIP premium, payout, and deductible mechanics; calibrated on a self-collected 937-household survey (Passaic River Basin, NJ) via structural equation modeling and Bayesian calibration; simulated payouts validated against OpenFEMA insurance-claims data across 27 census tracts over 13 years (2011–2023), reported over 50 Monte Carlo runs (median + IQR) rather than a single run. Research prototype, archived.

Catastrophe-risk modeling fluency: NFIP mechanics, exceedance-probability curves, coupled ABM × catastrophe modeling.

— EDUCATION

Ph.D., Civil & Environmental Engineering, Lehigh University	2024 – present · expected 2028
M.S., Hydrological and Oceanic Sciences, National Central University	Sep 2021 – Jun 2023
B.S., Earth Sciences, National Central University	Sep 2017 – Jun 2021

— SKILLS

AI-agent & research infrastructure: MCP servers & tooling · LLM-agent orchestration · multi-agent collaboration · agent governance & validation · research-workflow automation (Zotero / Obsidian / NotebookLM) · local model deployment (Ollama)

Programming & tooling: Python · testing & CI · Git · cross-platform (Windows / POSIX) engineering

Statistical modeling: structural equation modeling (SEM) · Bayesian calibration

Risk analytics: NFIP mechanics · exceedance-probability curves · coupled catastrophe modeling